

# Isha Chaudhary

ishachaudhary.web.illinois.edu

isha4@illinois.edu

ishcha

## Education

|                                                                                                                                             |                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>University of Illinois Urbana-Champaign</b><br><i>Ph.D. in Computer Science; Guide: Prof. Gagandeep Singh</i>                            | <b>2022 - 2027</b>                     |
| <b>University of Illinois Urbana-Champaign</b><br><i>M.S. in Computer Science with <a href="#">Thesis</a>; Guide: Prof. Gagandeep Singh</i> | <b>2022 - 2025</b><br>GPA: 3.97/4.00   |
| <b>Indian Institute of Technology Delhi, India</b><br><i>B.Tech in Electrical Engineering (minor in CS); GPA: 9.3/10.0</i>                  | <b>2018 - 2022</b><br>Class Rank: 1/51 |

## Research Interests

**Safety and trustworthiness of frontier AI systems:** certifying catastrophic risks and adversarial robustness with formal guarantees for LLM and agentic system behaviors

## Selected Publications

- **Isha Chaudhary**, Vedaant Jain, Gagandeep Singh. *QuaCer-C: Quantitative Certification of Knowledge Comprehension in LLMs*. **AISTATS 2026 Spotlight**, also in SeT LLM Workshop @ ICLR 2024.
- Chengxiao Wang, **Isha Chaudhary**, Qian Hu, Weitong Ruan, Rahul Gupta, Gagandeep Singh. *How Catastrophic is Your LLM? Certifying Risks in Conversation*. **ICLR 2026**.  
**Impact:** Deployed in Amazon's Nova evaluation suite
- **Isha Chaudhary**, Qian Hu, Manoj Kumar, Morteza Ziyadi, Rahul Gupta, Gagandeep Singh. *Quantitative Certification of Bias in Large Language Models*. **ICLR 2025**.
- **Isha Chaudhary**, Vedaant Jain, Avaljot Singh, Kavya Sachdeva, Sayan Ranu, Gagandeep Singh. *Lumos: Let there be Language Model System Certification*. Arxiv 2025.  
**Impact:** In deployment within National Research Platform's model cards
- Jason Vega\*, **Isha Chaudhary**\*, Changming Xu\*, Gagandeep Singh. *Bypassing the Safety Training of Open-Source LLMs with Priming Attacks*. **ICLR 2024** (Tiny Papers track). (\* indicates equal contribution)  
**Impact:** Attack Success Rate  $\uparrow 3.3\times$  over baselines on harmful behavior benchmarks
- **Isha Chaudhary**, Shuyi Lin, Cheng Tan, Gagandeep Singh. *Specification Generation for Neural Networks in Systems*. **ICSE 2026** (Poster), also in ML4Wireless Workshop @ ICML 2025.
- **Isha Chaudhary**, Alex Renda, Charith Mendis, Gagandeep Singh. *COMET: X86 Cost Model Explanation Framework*. **MLSys 2024**, also in XAI-in-Action Workshop @ NeurIPS 2023.

## Experience

|                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>Amazon AGI</b><br><i>Applied Scientist Intern with Dr. Rahul Gupta and Dr. Weitong Ruan</i>                                                                                                                                                                                                                                                                                                                                | <b>Boston, USA</b><br>May 2026 - Aug 2026       |
| <b>Amazon Web Services</b><br><i>Applied Scientist Intern with Dr. Ganyu Teng and Dr. Shawn Zhang</i>                                                                                                                                                                                                                                                                                                                         | <b>Seattle, USA</b><br>Sep 2025 - Dec 2025      |
| <ul style="list-style-type: none"><li>○ Developed a principled framework to stress-test scalable-oversight-enabled agentic systems via adversarial prompt injections, using CVaR to expose worst-case alignment failures hidden by average-case metrics</li><li>○ Finetuned Qwen-1.5B as a transferable adversary with GRPO to generate effective prompt-injection attacks against scalable-oversight judge models.</li></ul> |                                                 |
| <b>Microsoft Research</b><br><i>Research Intern with Dr. Ryan Beckett, Dr. Siva Kakarla, Prof. Francis Yan, Dr. Behnaz Arzani</i>                                                                                                                                                                                                                                                                                             | <b>Redmond, USA</b><br>May 2025 - Aug 2025      |
| Developed LLM-powered semantic analysis of system logs for anomaly detection, leveraging frontier model capabilities for safety-critical infrastructure monitoring.                                                                                                                                                                                                                                                           |                                                 |
| <b>Microsoft AI (Bing)</b><br><i>Data Science Intern</i>                                                                                                                                                                                                                                                                                                                                                                      | <b>Redmond, USA</b><br>May 2024 - Aug 2024      |
| Developed machine learning models for revenue optimization in Bing search.                                                                                                                                                                                                                                                                                                                                                    |                                                 |
| <b>Adobe Research (Cloud Tech group)</b><br><i>Research Intern / Guide: Dr. Gaurav Sinha / <a href="#">US patent no. US 2023/0274310 A1</a></i>                                                                                                                                                                                                                                                                               | <b>Bengaluru, India</b><br>May 2021 - Aug 2021  |
| Multi-task Learning with Label Proportions: Jointly disaggregated multiple aggregate labels with individual features                                                                                                                                                                                                                                                                                                          |                                                 |
| <b>Blockchain IoT Security Project: DST-JST Grant on IoT Security</b><br><i>Research Assistant / Guide: Prof. Subodh Sharma, IIT Delhi / Denso Intl. India</i>                                                                                                                                                                                                                                                                | <b>IIT Delhi, India</b><br>Sept 2020 - May 2021 |
| Designed & developed Hyperledger Fabric-based supply chain system for real-time order traceability & security                                                                                                                                                                                                                                                                                                                 |                                                 |

## Awards

- o Awarded **Best Presentation Award** (Security and Privacy track) at CSL student conference 2025 (2025)
- o Awarded **Institute Silver Medal & Prof. C.S. Jha Memorial Excellence Award** at IIT Delhi (2022)
- o **Top 7% merit award**: In top 7% high GPA students at IIT Delhi for 5 semesters: semesters 3, 4, 6, 7, 8 (2019-22)
- o **Design & Innovation Summer Award (MHRD Grant, Govt. of India)** (2019)
- o **KVPY**: Qualified for the Kishore Vaigyanik Protsahan Yojana Fellowship (IISc); All India Rank 232 (2016)
- o **NTSE**: Received National Talent Search Scholarship, given to top 1000 students nationwide by NCERT (2015)

## Talks

### Certifying LLMs

- o NSF's National AI Research Resource (NAIRR) Pilot 2026
- o Microsoft Research Redmond and Microsoft Research India 2025
- o CSL student conference 2025
- o IIT Delhi 2025
- o Amazon-Illinois Center on AI for Interactive Conversational Experiences 2024

### Teaching Smarter with AI: How to Use AI in Classrooms Without Losing Control

- o Graduate TA training academy organized by CITL, UIUC 2025

### Exploring COMET: Neural Cost Model Explanation Framework

- o Google Compiler Research reading group 2024
- o VMWare Research 2024

## Teaching

### CS 477: Formal Software Development Methods

UIUC, USA

*Guest Lectures on formally certifying frontier models*

*Apr 2026*

### Grad Academy, CITL

UIUC, USA

*Instructor: Isha Chaudhary, UIUC*

*Aug 19, 2025*

### Advanced Topics in Programming Systems - Trustworthy AI Systems (CS521)

UIUC, USA

*Instructor: Prof. Gagandeep Singh, UIUC*

*Aug 2024 - Dec 2024*

### Calculus (MTL100)

IIT Delhi, India

*Instructor: Prof. Sivananthan Sampath, IIT Delhi*

*Aug 2021 - Dec 2021*

## Grants

- o Awarded **ICML 2023 Travel Grant** (2023)
- o Awarded scholarship for **Programming Languages Mentoring Workshop @ PLDI 2023** (2023)
- o **NSF Travel Grant for IEEE SaTML 2023** (2022)

## Service

- o Area Chair for MLSys 2025.
- o Reviewer for ICML 2026; KDD2026; COLM 2026; ICLR 2025, 2026; FAccT 2025, 2026; MLSys 2025; AIES 2024, 2025; Tiny Papers @ ICLR 2024.
- o Artifact Evaluator for VMCAI 2024.
- o Main organizer for Formal Methods and Software Engineering seminar, Spring 2024 & Fall 2024, at UIUC.
- o Mentor for the Undergraduate Research Apprenticeship Program (URAP) at UIUC, 2024-25
- o Mentor in the Undergraduate Research in Scientific Advancement (URSA) program at UIUC, 2024-25
- o Mentor in SIGPLAN-M long-term mentorship programme
- o Mentor in K-12 Broadening Participation in Computing program at UIUC, Summer 2023.  
(Guided a team of high-school students in successfully learning & conducting research on LLMs.)

## Undergraduate Thesis Project

### Security & Privacy of SCADA systems & Microgrids

IIT Delhi, India

*Guides: Prof. Anupam Joshi & Prof. B.K. Panigrahi*

*Aug 2021 - Dec 2021*

- o Developed Multi-task data-driven real-time Intrusion detection system for digital twin of AC Microgrid (AUC: 0.85)